NAMA PEMILIK PROYEK : **PT PERTAMINA EP**

NAMA PROYEK : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

LOKASI : **SP TANJUNG MIRING BARAT & KARANGAN TIMUR - FIELD LIMAU**

NO. KONTRAK : **4710002197**

REV.	TGL.	DESKRIPSI	DISIAPKAN	DIPERIKSA	DISETUJUI	DIPERIKSA	DISETUJUI
A	14/01/2025	Issued For Review	FN	AS	MR		
			PT. PROSYMPAC OIL & GAS			PT. PERTAMINA EP	

TABEL REVISI

Rev.	Tanggal	Deskripsi

COMMENT RESPONSE SHEET

No.	Page	Company's Comment	Contractor's Response	Remarks

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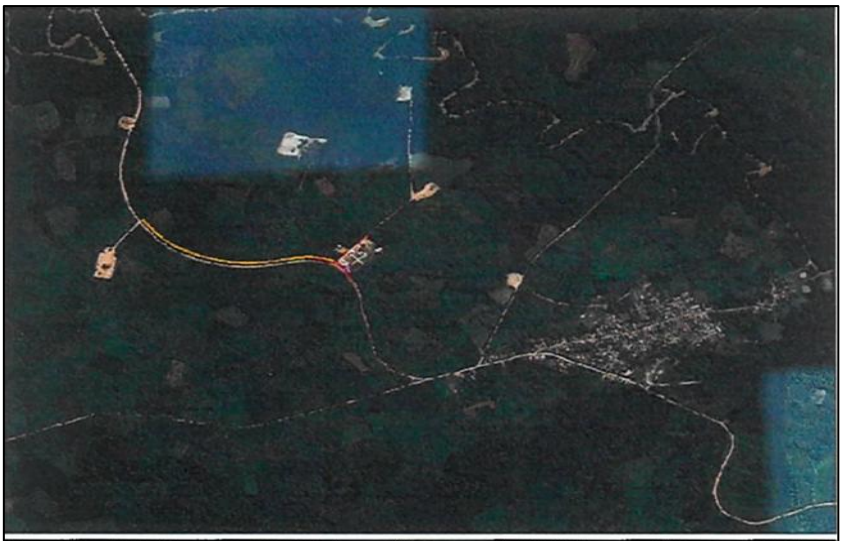
1. GENERAL

1.1 Latar Belakang

Struktur Tanjung Miring Barat (TMB) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 20 kilometer di sebelah tenggara Kota Prabumulih, dengan luas daerah sekitar 8 km². Struktur Karangan (KRG) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 15 kilometer di sebelah selatan Kota Prabumulih, dengan luas daerah sekitar 45 km².



ID	Easting (Meter)	Northing (Meter)	Elevation (Meter)	Feature Code
PN-1531	424962,745	96 7376,210	10,414	
PM-8593	425000,093	907528.777	19,046	

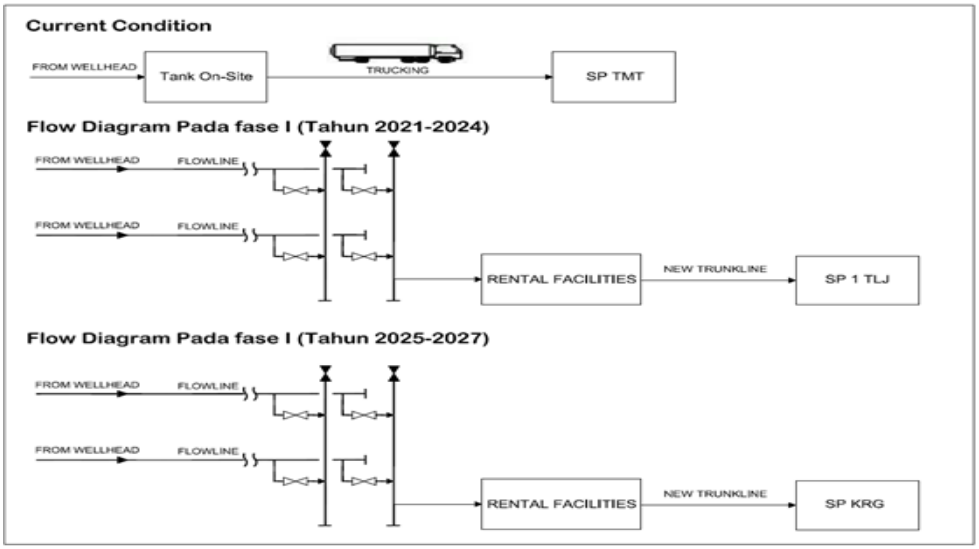
Gambar 1. 1 Lokasi Rencana SP Tanjung Miring Barat – Karangan Timur

Lapangan Tanjung Miring Barat telah diproduksi sejak tahun 1938 dan terus beroperasi hingga tahun 2003 dengan produksi terakhir dari sumur TMB-11. Sejak dinonaktifkannya lapangan ini, fasilitas produksi yang terdiri dari *flowline*, stasiun pengumpul, serta jalur pengiriman pipa minyak ke custody transfer (PPP Prabumulih) sudah tidak layak operasi. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Baral serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah. Saat ini status SP TMB adalah *idle* atau tidak beroperasi, sehingga produksi sumuran menggunakan *Tank On-Site* (TOS) dan dikirim secara gross ke SP-II TMT menggunakan *trucking*. Dikarenakan tidak adanya line gas ke sumur-sumur struktur TMB, supply Listrik untuk pumping unit menggunakan solar (diesel engine). Lapangan Karangan Timur hanya diproduksikan dari 2 sumur yaitu KRG-9 dan KRG-10 beroperasi mulai tahun 1984, dimana kondisi eksisting jalur pengiriman minyak dari sumur menggunakan *trucking*. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Barat serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah.

Dalam rencana jangka panjang sebagai upaya peningkatan produksi, PT Pertamina EP berencana akan mengembangkan Lapangan Tanjung Miring Barat dan Karangan Timur melalui

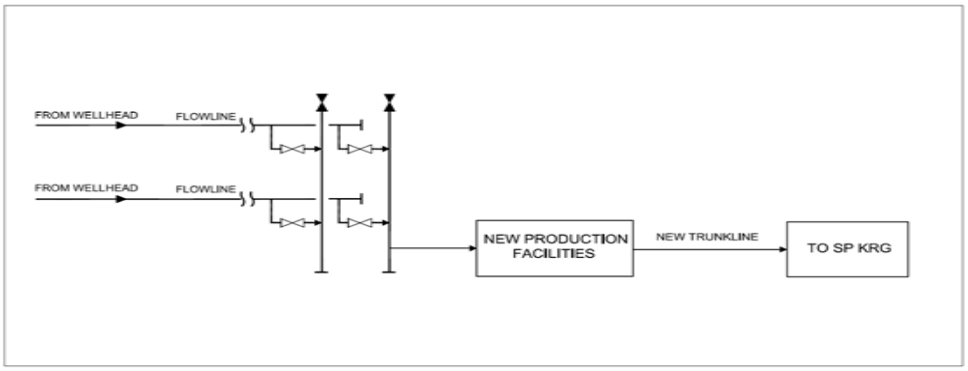
pengembangan fasilitas produksi dimana rencana kerja pengembangan fasilitas akan dibagi ke dalam dua fase.

Pada fase pertama, produksi dari pengembangan awal TMB -KRG Timur akan dikirimkan menuju fasilitas sewa *Early Production Facility* (EPF) untuk selanjutnya dikirimkan menuju SP- 1 TLJ melalui *trunkline* baru. Pada fase ini lingkup fasilitas produksi yang akan dibangun adalah *flowline* dari sumuran, sewa EPF dan trunkline dari EPF sampai dengan SP-1 TLJ.



Gambar 1. 2 Rencana Pola Operasi Fase

Pada fase kedua, akan dilakukan pengembangan tambahan dari sumur-sumur TMB-KRG Timur. Pada fase ini, lingkup fasilitas produksi yang akan dibangun meliputi pembangunan flowline dari sumuran pengembangan lanjutan serta peningkatan fasilitas SP TMB untuk menggantikan sewa EPF.



Gambar 1. 3 Rencana Pola Operasi Fase

1.2 Tujuan

Tujuan dari dokumen ini adalah untuk untuk menentukan kebutuhan dan parameter sistem proteksi katodik yang digunakan pada struktur atau peralatan logam, guna mencegah terjadinya korosi akibat reaksi elektrokimia dengan lingkungan sekitarnya. Perhitungan dilakukan untuk memastikan bahwa sistem proteksi katodik yang dirancang mampu memberikan tingkat perlindungan yang memadai sesuai dengan kriteria teknis dan standar yang berlaku, sehingga

umur operasional peralatan dapat ditingkatkan serta risiko kegagalan akibat korosi dapat diminimalkan.

1.3 Ruang Lingkup

Ruang lingkup dokumen *Cathodic Protection Calculation* ini mencakup perhitungan teknis sistem proteksi katodik berdasarkan metode dan parameter yang tercantum dalam lembar perhitungan Excel, termasuk penentuan luas permukaan struktur yang dilindungi, kebutuhan arus proteksi, pemilihan jenis dan jumlah anoda, serta estimasi umur desain sistem proteksi katodik. Perhitungan dilakukan dengan mempertimbangkan kondisi lingkungan, karakteristik material, dan asumsi desain yang digunakan, sehingga hasil perhitungan dapat dijadikan sebagai dasar perencanaan, implementasi, dan evaluasi sistem *cathodic protection* pada proyek terkait.

1.4 Istilah

Dokumen ini berisi istilah-istilah yang memiliki definisi tercantum sebagai berikut:

Tabel 1. 1 Daftar Istilah

Perusahaan	Perusahaan yang dimaksud adalah PT PERTAMINA HULU ROKAN selaku Pemilik dan Pengguna Fasilitas akhir.
Kontraktor	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan pengadaan dan konstruksi yang akan bertanggung jawab kepada PERUSAHAAN.
Vendor	Pihak lain yang akan menyediakan sesuatu untuk kebutuhan proyek, selain yang disediakan oleh PERUSAHAAN dan KONTRAKTOR, disebut VENDOR. Tanggung jawabnya langsung kepada KONTRAKTOR baik terhadap PO maupun bentuk perjanjian lain antara VENDOR dan KONTRAKTOR.
SKK MIGAS	Suatu badan dari Pemerintah Indonesia yang bertanggung jawab terhadap proses persetujuan dan izin untuk semua fasilitas di lingkungan minyak dan gas.
Perencana	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan Jasa DED Engineering Services. Dalam hal ini disebut PERENCANA, yang akan bertanggung jawab kepada PERUSAHAAN sebagaimana dirinci dalam kontrak, scope of work, atau work order.

Inspektor	Perwakilan dari PERUSAHAAN dan/atau seseorang yang ditunjuk oleh PERUSAHAAN, yang bertanggung jawab sebagai INSPEKTOR atas pekerjaan yang dilakukan oleh KONTRAKTOR.SINGKATAN

2. KODE, STANDAR, DAN REFERENSI

Tabel 2. 1 International Code and Standard

No	Referensi	Deskripsi
1	IEEE 399	IEEE Recommended Practice for Industrial and Commercial Power Systems Analysis
2	NFPA 70 / NEC	National Electrical Code
3	NACE RP0176	Corrosion Control of Steel Fixed Offshore Platforms Associated with Petroleum Production
4	NACE RP0575	Internal Cathodic Protection Systems for Steel Water Pipelines
5	API Recommended Practice 651	Cathodic Protection of Aboveground Petroleum Storage Tanks

Tabel 2. 2 Dokumen Proyek

No.	No. Dokumen		Nama Dokumen
1	TMBP-SP001-EL-DBS-D101	-	Electrical Design Criteria
2	TMBP-SP001-EL-SPE-D102	-	Specification for Cathodic Protection
3	TMBP-SP001-EL-SPE-D103	-	Specification For Electrical Installation & Testing
4	TMBP-01-PR-PID-D103	-	Piping and Instrumentation Diagram for Test Tank TMB (T-003) & Test Pump TMB (P-002A)
5	TMBP-01-PR-PID-D104	-	Piping and Instrumentation Diagram for Test Tank KRG (T-004) & Test Pump KRG (P-003 A/B)
6	TMBP-01-PR-PID-D105	-	Piping and Instrumentation Diagram for Storage Tank TMB (T-005 A/B)
7	TMBP-01-PR-PID-D106	-	Piping and Instrumentation Diagram for Storage Tank KRG (T-006 AB)
8	TMBP-01-PR-PID-D108	-	Piping and Instrumentation Diagram for Produced Water Tank (T-007) and Feed Water Pump (P-004 A/B)

LAMPIRAN

ELECTRICAL CATHODIC PROTECTION CALCULATION

DESIGN OF CATHODIC PROTECTION SYSTEM
FOR WASH TANK TMB (T-001)

DESIGN CRITERIA		
1 System Type	Sacrificial Anode	
2 Anode Material	Magnesium Anode High Potential 32 lbs	
3 Weight of Anode (ma)	14.5	kg
4 Length of Anode (L_a)	1.346	m
5 Diameter Anode (r_a)	0.165	m
6 Anode current capacity (ε)	1,100	Ah/kg / NACE RP0575 clause 4.2
7 Anode utilisation factor (0<μ<1) (U)	0.85	Lampiran 2
8 Natural potential (Va)	-1.75	V / Table-2 API RP 651 clause 6.1
9 Protective potential (Vp)	-0.85	V / API RP 651 clause 8.2.2.2
10 Coating Breakdown factor (fc)	0.18	DNV-RP-B401 clause 6.4.4
11 Design Current Density (i_c)	55	mA/m² / NACE RP0176 table A1
12 Design Life (tf)	20	tahun
13 Sendiment / Soil Resistivity (ρ)	112	Ohm.m / TMBP-01-A0-REP-501
TANK DATA		
1 Diameter (D)	7.32	(meter)
2 Height (H)	7.32	(meter)
3 Assumption water level (from bottom)	0.60	(meter)
DESIGN CALCULATIONS		
1 Tank surface area (eq. 1)		
$S_{total} = \pi \times r^2 + 2\pi r \times h$	55.81	m²
2 Current protection requirement (eq. 2)		
$I_c = \frac{i_c}{1000} \times A \times fc$	0.553	Ampere
3 Calculating Anode Resistance (eq. 3)		
$R_a = \frac{\rho}{2\pi L_a}[\ln\left(\frac{4L_a}{r_a}\right) - 1]$	32.91	Ohm
4 Calculating Output current (eq. 4)		
$I_a = \frac{V_p - V_a}{R_a}$	0.03	Ampere
5 Total net Anode mass (eq. 5)		
$M_a = \frac{I_c \times tf \times 8760}{\mu \times \varepsilon}$	103.54	kg
6 Calculating Min. Number of Anodes due to Mass (eq. 6)		
$N_{min,m} = \frac{M_a}{m_a}$	7	pcs
7 Calculating Min. Number of Anodes due to Current (eq. 7)		
$N_{min,i} = \frac{I_c}{I_a}$	20	pcs
8 Selected Number of Anodes (eq. 8)		
$N = \text{Max}\,[N_{min,m}, N_{min,i}]$	20	pcs
9 Individual Anode Current Capacity (eq. 9)		
$C_a = m_a \times \varepsilon \times \mu$	13,558	Ah
10 Crosscheck to DNV RP B401 (eq. 10)		
$N \times C_a$	273,956	
$I_c \times tf \times 8760$	96,809	
$N \times C_a \geq I_c \times tf \times 8760$	**OK**	

DESIGN OF CATHODIC PROTECTION SYSTEM

FOR WASH TANK KRG (T-002)

DESIGN CRITERIA		
1 System Type	Sacrificial Anode	
2 Anode Material	Magnesium Anode High Potential 32 lbs	
3 Weight of Anode (ma)	14.5	kg
4 Length of Anode (L_a)	1.346	m
5 Diameter Anode (r_a)	0.165	m
6 Anode current capacity (ε)	1,100	Ah/kg / NACE RP0575 clause 4.2
7 Anode utilisation factor (0<μ<1) (U)	0.85	Lampiran 2
8 Natural potential (Va)	-1.75	V / Table-2 API RP 651 clause 6.1
9 Protective potential (Vp)	-0.85	V / API RP 651 clause 8.2.2.2
10 Coating Breakdown factor (fc)	0.18	DNV-RP-B401 clause 6.4.4
11 Design Current Density (i_c)	55	mA/m² / NACE RP0176 table A1
12 Design Life (tf)	20	tahun
13 Sendiment / Soil Resistivity (ρ)	112	Ohm.m / TMBP-01-A0-REP-501
TANK DATA		
1 Diameter (D)	9.15	(meter)
2 Height (H)	7.32	(meter)
3 Assumption water level (from bottom)	0.60	(meter)
DESIGN CALCULATIONS		
1 Tank surface area (eq. 1)		
$S_{total} = \pi \times r^2 + 2\pi r \times h$	83.04	m²
2 Current protection requirement (eq. 2)		
$I_c = \frac{i_c}{1000} \times A \times fc$	0.822	Ampere
3 Calculating Anode Resistance (eq. 3)		
$R_a = \frac{\rho}{2\pi L_a}[\ln\left(\frac{4L_a}{r_a}\right) - 1]$	32.91	Ohm
4 Calculating Output current (eq. 4)		
$I_a = \frac{V_p - V_a}{R_a}$	0.03	Ampere
5 Total net Anode mass (eq. 5)		
$M_a = \frac{I_c \times tf \times 8760}{\mu \times \varepsilon}$	154.04	kg
6 Calculating Min. Number of Anodes due to Mass (eq. 6)		
$N_{min,m} = \frac{M_a}{m_a}$	11	pcs
7 Calculating Min. Number of Anodes due to Current (eq. 7)		
$N_{min,i} = \frac{I_c}{I_a}$	30	pcs
8 Selected Number of Anodes (eq. 8)		
$N = \text{Max}\ [N_{min,m}, N_{min,i}]$	30	pcs
9 Individual Anode Current Capacity (eq. 9)		
$C_a = m_a \times \varepsilon \times \mu$	13,558	Ah
10 Crosscheck to DNV RP B401 (eq. 10)		
$N \times C_a$	407,565	
$I_c \times tf \times 8760$	144,023	
$N \times C_a \geq I_c \times tf \times 8760$	**OK**	

Note : This calculation shall be confirmed by VENDOR during detail engineering

DESIGN OF CATHODIC PROTECTION SYSTEM

FOR TEST TANK TMB (T-003)

DESIGN CRITERIA		
1 System Type	Sacrificial Anode	
2 Anode Material	Magnesium Anode High Potential 32 lbs	
3 Weight of Anode (ma)	14.5	kg
4 Length of Anode (L_a)	1.346	m
5 Diameter Anode (r_a)	0.165	m
6 Anode current capacity (ε)	1,100	Ah/kg / NACE RP0575 clause 4.2
7 Anode utilisation factor (0<μ<1) (U)	0.85	Lampiran 2
8 Natural potential (Va)	-1.75	V / Table-2 API RP 651 clause 6.1
9 Protective potential (Vp)	-0.85	V / API RP 651 clause 8.2.2.2
10 Coating Breakdown factor (fc)	0.18	DNV-RP-B401 clause 6.4.4
11 Design Current Density (i_c)	55	mA/m² / NACE RP0176 table A1
12 Design Life (tf)	20	tahun
13 Sendiment / Soil Resistivity (ρ)	112	Ohm.m / TMBP-01-A0-REP-501
TANK DATA		
1 Diameter (D)	4.27	(meter)
2 Height (H)	3.65	(meter)
3 Assumption water level (from bottom)	0.60	(meter)
DESIGN CALCULATIONS		
1 Tank surface area (eq. 1)		
$S_{total} = \pi \times r^2 + 2\pi r \times h$	22.34	m²
2 Current protection requirement (eq. 2)		
$I_c = \frac{i_c}{1000} \times A \times fc$	0.221	Ampere
3 Calculating Anode Resistance (eq. 3)		
$R_a = \frac{\rho}{2\pi L_a}[\ln\left(\frac{4L_a}{r_a}\right) - 1]$	32.91	Ohm
4 Calculating Output current (eq. 4)		
$I_a = \frac{V_p - V_a}{R_a}$	0.03	Ampere
5 Total net Anode mass (eq. 5)		
$M_a = \frac{I_c \times tf \times 8760}{\mu \times \varepsilon}$	41.45	kg
6 Calculating Min. Number of Anodes due to Mass (eq. 6)		
$N_{min,m} = \frac{M_a}{m_a}$	3	pcs
7 Calculating Min. Number of Anodes due to Current (eq. 7)		
$N_{min,i} = \frac{I_c}{I_a}$	8	pcs
8 Selected Number of Anodes (eq. 8)		
$N = \mathrm{Max}\,[N_{min,m}, N_{min,i}]$	8	pcs
9 Individual Anode Current Capacity (eq. 9)		
$C_a = m_a \times \varepsilon \times \mu$	13,558	Ah
10 Crosscheck to DNV RP B401 (eq. 10)		
$N \times C_a$	109,667	
$I_c \times tf \times 8760$	38,754	
$N \times C_a \geq I_c \times tf \times 8760$	**OK**	

Note : This calculation shall be confirmed by VENDOR during detail engineering

DESIGN OF CATHODIC PROTECTION SYSTEM
FOR TEST TANK KRG (T-004)

DESIGN CRITERIA		
1 System Type	Sacrificial Anode	
2 Anode Material	Magnesium Anode High Potential 32 lbs	
3 Weight of Anode (ma)	14.5	kg
4 Length of Anode (L_a)	1.346	m
5 Diameter Anode (r_a)	0.165	m
6 Anode current capacity (Ɛ)	1,100	Ah/kg / NACE RP0575 clause 4.2
7 Anode utilisation factor (0<μ<1) (U)	0.85	Lampiran 2
8 Natural potential (Va)	-1.75	V / Table-2 API RP 651 clause 6.1
9 Protective potential (Vp)	-0.85	V / API RP 651 clause 8.2.2.2
10 Coating Breakdown factor (fc)	0.18	DNV-RP-B401 clause 6.4.4
11 Design Current Density (i_c)	55	mA/m² / NACE RP0176 table A1
12 Design Life (tf)	20	tahun
13 Sendiment / Soil Resistivity (ρ)	112	Ohm.m / TMBP-01-A0-REP-501
TANK DATA		
1 Diameter (D)	5.49	(meter)
2 Height (H)	3.65	(meter)
3 Assumption water level (from bottom)	0.60	(meter)
DESIGN CALCULATIONS		
1 Tank surface area (eq. 1)		
$S_{total} = \pi \times r^2 + 2\pi r \times h$	34.06	m²
2 Current protection requirement (eq. 2)		
$I_c = \frac{i_c}{1000} \times A \times fc$	0.337	Ampere
3 Calculating Anode Resistance (eq. 3)		
$R_a = \frac{\rho}{2\pi L_a}[\ln\left(\frac{4L_a}{r_a}\right) - 1]$	32.91	Ohm
4 Calculating Output current (eq. 4)		
$I_a = \frac{V_p - V_a}{R_a}$	0.03	Ampere
5 Total net Anode mass (eq. 5)		
$M_a = \frac{I_c \times tf \times 8760}{\mu \times \varepsilon}$	63.19	kg
6 Calculating Min. Number of Anodes due to Mass (eq. 6)		
$N_{min,m} = \frac{M_a}{m_a}$	4	pcs
7 Calculating Min. Number of Anodes due to Current (eq. 7)		
$N_{min,i} = \frac{I_c}{I_a}$	12	pcs
8 Selected Number of Anodes (eq. 8)		
$N = \text{Max}\ [N_{min,m}, N_{min,i}]$	12	pcs
9 Individual Anode Current Capacity (eq. 9)		
$C_a = m_a \times \varepsilon \times \mu$	13,558	Ah
10 Crosscheck to DNV RP B401 (eq. 10)		
$N \times C_a$	167,190	
$I_c \times tf \times 8760$	59,081	
$N \times C_a \geq I_c \times tf \times 8760$	**OK**	

Note : This calculation shall be confirmed by VENDOR during detail engineering

DESIGN OF CATHODIC PROTECTION SYSTEM

FOR PRODUCED WATER TANK (T-007)

DESIGN CRITERIA		
1 System Type	Sacrificial Anode	
2 Anode Material	Magnesium Anode High Potential 32 lbs	
3 Weight of Anode (ma)	14.5	kg
4 Length of Anode (L_a)	1.346	m
5 Diameter Anode (r_a)	0.165	m
6 Anode current capacity (ε)	1,100	Ah/kg / NACE RP0575 clause 4.2
7 Anode utilisation factor (0<μ<1) (U)	0.85	Lampiran 2
8 Natural potential (Va)	-1.75	V / Table-2 API RP 651 clause 6.1
9 Protective potential (Vp)	-0.85	V / API RP 651 clause 8.2.2.2
10 Coating Breakdown factor (fc)	0.18	DNV-RP-B401 clause 6.4.4
11 Design Current Density (i_c)	55	mA/m² / NACE RP0176 table A1
12 Design Life (tf)	20	tahun
13 Sendiment / Soil Resistivity (ρ)	112	Ohm.m / TMBP-01-A0-REP-501
TANK DATA		
1 Diameter (D)	7.32	(meter)
2 Height (H)	4.88	(meter)
3 Assumption water level (from bottom)	0.60	(meter)
DESIGN CALCULATIONS		
1 Tank surface area (eq. 1)		
$S_{total} = \pi \times r^2 + 2\pi r \times h$	55.81	m²
2 Current protection requirement (eq. 2)		
$I_c = \frac{i_c}{1000} \times A \times fc$	0.553	Ampere
3 Calculating Anode Resistance (eq. 3)		
$R_a = \frac{\rho}{2\pi L_a}[\ln\left(\frac{4L_a}{r_a}\right) - 1]$	32.91	Ohm
4 Calculating Output current (eq. 4)		
$I_a = \frac{V_p - V_a}{R_a}$	0.03	Ampere
5 Total net Anode mass (eq. 5)		
$M_a = \frac{I_c \times tf \times 8760}{\mu \times \varepsilon}$	103.54	kg
6 Calculating Min. Number of Anodes due to Mass (eq. 6)		
$N_{min,m} = \frac{M_a}{m_a}$	7	pcs
7 Calculating Min. Number of Anodes due to Current (eq. 7)		
$N_{min,i} = \frac{I_c}{I_a}$	20	pcs
8 Selected Number of Anodes (eq. 8)		
$N = \text{Max}\ [N_{min,m}, N_{min,i}]$	20	pcs
9 Individual Anode Current Capacity (eq. 9)		
$C_a = m_a \times \varepsilon \times \mu$	13,558	Ah
10 Crosscheck to DNV RP B401 (eq. 10)		
$N \times C_a$	273,956	
$I_c \times tf \times 8760$	96,809	
$N \times C_a \geq I_c \times tf \times 8760$	**OK**	

Note : This calculation shall be confirmed by VENDOR during detail engineering

DESIGN OF CATHODIC PROTECTION SYSTEM
FOR TREATED WATER TANK (T-008)

DESIGN CRITERIA		
1 System Type	Sacrificial Anode	
2 Anode Material	Magnesium Anode High Potential 32 lbs	
3 Weight of Anode (ma)	14.5	kg
4 Length of Anode (L_a)	1.346	m
5 Diameter Anode (r_a)	0.165	m
6 Anode current capacity (ε)	1,100	Ah/kg / NACE RP0575 clause 4.2
7 Anode utilisation factor (0<μ<1) (U)	0.85	Lampiran 2
8 Natural potential (Va)	-1.75	V / Table-2 API RP 651 clause 6.1
9 Protective potential (Vp)	-0.85	V / API RP 651 clause 8.2.2.2
10 Coating Breakdown factor (fc)	0.18	DNV-RP-B401 clause 6.4.4
11 Design Current Density (i_c)	55	mA/m² / NACE RP0176 table A1
12 Design Life (tf)	20	tahun
13 Sendiment / Soil Resistivity (ρ)	112	Ohm.m / TMBP-01-A0-REP-501
TANK DATA		
1 Diameter (D)	7.32	(meter)
2 Height (H)	4.88	(meter)
3 Assumption water level (from bottom)	0.60	(meter)
DESIGN CALCULATIONS		
1 Tank surface area (eq. 1)		
$S_{total} = \pi \times r^2 + 2\pi r \times h$	55.81	m²
2 Current protection requirement (eq. 2)		
$I_c = \frac{i_c}{1000} \times A \times fc$	0.553	Ampere
3 Calculating Anode Resistance (eq. 3)		
$R_a = \frac{\rho}{2\pi L_a}[\ln\left(\frac{4L_a}{r_a}\right) - 1]$	32.91	Ohm
4 Calculating Output current (eq. 4)		
$I_a = \frac{V_p - V_a}{R_a}$	0.03	Ampere
5 Total net Anode mass (eq. 5)		
$M_a = \frac{I_c \times tf \times 8760}{\mu \times \varepsilon}$	103.54	kg
6 Calculating Min. Number of Anodes due to Mass (eq. 6)		
$N_{min,m} = \frac{M_a}{m_a}$	7	pcs
7 Calculating Min. Number of Anodes due to Current (eq. 7)		
$N_{min,i} = \frac{I_c}{I_a}$	20	pcs
8 Selected Number of Anodes (eq. 8)		
$N = \text{Max}\ [N_{min,m}, N_{min,i}]$	20	pcs
9 Individual Anode Current Capacity (eq. 9)		
$C_a = m_a \times \varepsilon \times \mu$	13,558	Ah
10 Crosscheck to DNV RP B401 (eq. 10)		
$N \times C_a$	273,956	
$I_c \times tf \times 8760$	96,809	
$N \times C_a \geq I_c \times tf \times 8760$	**OK**	

DESIGN OF CATHODIC PROTECTION SYSTEM
FOR STORAGE WATER TANK KRG (T-006 B)

DESIGN CRITERIA		
1 System Type	Sacrificial Anode	
2 Anode Material	Magnesium Anode High Potential 32 lbs	
3 Weight of Anode (ma)	14.5	kg
4 Length of Anode (L_a)	1.346	m
5 Diameter Anode (r_a)	0.165	m
6 Anode current capacity (Ɛ)	1,100	Ah/kg / NACE RP0575 clause 4.2
7 Anode utilisation factor (0<μ<1) (U)	0.85	Lampiran 2
8 Natural potential (Va)	-1.75	V / Table-2 API RP 651 clause 6.1
9 Protective potential (Vp)	-0.85	V / API RP 651 clause 8.2.2.2
10 Coating Breakdown factor (fc)	0.18	DNV-RP-B401 clause 6.4.4
11 Design Current Density (i_c)	55	mA/m² / NACE RP0176 table A1
12 Design Life (tf)	20	tahun
13 Sendiment / Soil Resistivity (ρ)	112	Ohm.m / TMBP-01-A0-REP-501
TANK DATA		
1 Diameter (D)	9.45	(meter)
2 Height (H)	4.88	(meter)
3 Assumption water level (from bottom)	0.60	(meter)
DESIGN CALCULATIONS		
1 Tank surface area (eq. 1)		
$S_{total} = \pi \times r^2 + 2\pi r \times h$	87.95	m²
2 Current protection requirement (eq. 2)		
$I_c = \frac{i_c}{1000} \times A \times fc$	0.871	Ampere
3 Calculating Anode Resistance (eq. 3)		
$R_a = \frac{\rho}{2\pi L_a}[\ln\left(\frac{4L_a}{r_a}\right) - 1]$	32.91	Ohm
4 Calculating Output current (eq. 4)		
$I_a = \frac{V_p - V_a}{R_a}$	0.03	Ampere
5 Total net Anode mass (eq. 5)		
$M_a = \frac{I_c \times tf \times 8760}{\mu \times \varepsilon}$	163.15	kg
6 Calculating Min. Number of Anodes due to Mass (eq. 6)		
$N_{min,m} = \frac{M_a}{m_a}$	11	pcs
7 Calculating Min. Number of Anodes due to Current (eq. 7)		
$N_{min,i} = \frac{I_c}{I_a}$	32	pcs
8 Selected Number of Anodes (eq. 8)		
$N = \text{Max}\ [N_{min,m}, N_{min,i}]$	32	pcs
9 Individual Anode Current Capacity (eq. 9)		
$C_a = m_a \times \varepsilon \times \mu$	13,558	Ah
10 Crosscheck to DNV RP B401 (eq. 10)		
$N \times C_a$	431,692	
$I_c \times tf \times 8760$	152,549	
$N \times C_a \geq I_c \times tf \times 8760$	**OK**	

Note : This calculation shall be confirmed by VENDOR during detail engineering

DESIGN OF CATHODIC PROTECTION SYSTEM
FOR STORAGE WATER TANK TMB (T-005 A)

DESIGN CRITERIA		
1 System Type	Sacrificial Anode	
2 Anode Material	Magnesium Anode High Potential 32 lbs	
3 Weight of Anode (ma)	14.5	kg
4 Length of Anode (L_a)	1.346	m
5 Diameter Anode (r_a)	0.165	m
6 Anode current capacity (Ɛ)	1,100	Ah/kg / NACE RP0575 clause 4.2
7 Anode utilisation factor (0<μ<1) (U)	0.85	Lampiran 2
8 Natural potential (Va)	-1.75	V / Table-2 API RP 651 clause 6.1
9 Protective potential (Vp)	-0.85	V / API RP 651 clause 8.2.2.2
10 Coating Breakdown factor (fc)	0.18	DNV-RP-B401 clause 6.4.4
11 Design Current Density (i_c)	55	mA/m² / NACE RP0176 table A1
12 Design Life (tf)	20	tahun
13 Sendiment / Soil Resistivity (ρ)	112	Ohm.m / TMBP-01-A0-REP-501
TANK DATA		
1 Diameter (D)	4.27	(meter)
2 Height (H)	4.88	(meter)
3 Assumption water level (from bottom)	0.60	(meter)
DESIGN CALCULATIONS		
1 Tank surface area (eq. 1)		
$S_{total} = \pi \times r^2 + 2\pi r \times h$	22.37	m²
2 Current protection requirement (eq. 2)		
$I_c = \frac{i_c}{1000} \times A \times fc$	0.221	Ampere
3 Calculating Anode Resistance (eq. 3)		
$R_a = \frac{\rho}{2\pi L_a}[\ln\left(\frac{4L_a}{r_a}\right) - 1]$	32.91	Ohm
4 Calculating Output current (eq. 4)		
$I_a = \frac{V_p - V_a}{R_a}$	0.03	Ampere
5 Total net Anode mass (eq. 5)		
$M_a = \frac{I_c \times tf \times 8760}{\mu \times \varepsilon}$	41.50	kg
6 Calculating Min. Number of Anodes due to Mass (eq. 6)		
$N_{min,m} = \frac{M_a}{m_a}$	3	pcs
7 Calculating Min. Number of Anodes due to Current (eq. 7)		
$N_{min,i} = \frac{I_c}{I_a}$	8	pcs
8 Selected Number of Anodes (eq. 8)		
$N = \text{Max}\ [N_{min,m}, N_{min,i}]$	8	pcs
9 Individual Anode Current Capacity (eq. 9)		
$C_a = m_a \times \varepsilon \times \mu$	13,558	Ah
10 Crosscheck to DNV RP B401 (eq. 10)		
$N \times C_a$	109,794	
$I_c \times tf \times 8760$	38,798	
$N \times C_a \geq I_c \times tf \times 8760$	**OK**	

Note : This calculation shall be confirmed by VENDOR during detail engineering

DESIGN OF CATHODIC PROTECTION SYSTEM

FOR STORAGE WATER TANK TMB (T-005 B)

DESIGN CRITERIA		
1 System Type	Sacrificial Anode	
2 Anode Material	Magnesium Anode High Potential 32 lbs	
3 Weight of Anode (ma)	14.5	kg
4 Length of Anode (L_a)	1.346	m
5 Diameter Anode (r_a)	0.165	m
6 Anode current capacity (Ɛ)	1,100	Ah/kg / NACE RP0575 clause 4.2
7 Anode utilisation factor (0<μ<1) (U)	0.85	Lampiran 2
8 Natural potential (Va)	-1.75	V / Table-2 API RP 651 clause 6.1
9 Protective potential (Vp)	-0.85	V / API RP 651 clause 8.2.2.2
10 Coating Breakdown factor (fc)	0.18	DNV-RP-B401 clause 6.4.4
11 Design Current Density (i_c)	55	mA/m² / NACE RP0176 table A1
12 Design Life (tf)	20	tahun
13 Sendiment / Soil Resistivity (ρ)	112	Ohm.m / TMBP-01-A0-REP-501
TANK DATA		
1 Diameter (D)	4.27	(meter)
2 Height (H)	4.88	(meter)
3 Assumption water level (from bottom)	0.60	(meter)
DESIGN CALCULATIONS		
1 Tank surface area (eq. 1)		
$S_{total} = \pi \times r^2 + 2\pi r \times h$	22.37	m²
2 Current protection requirement (eq. 2)		
$I_c = \frac{i_c}{1000} \times A \times fc$	0.221	Ampere
3 Calculating Anode Resistance (eq. 3)		
$R_a = \frac{\rho}{2\pi L_a}[\ln\left(\frac{4L_a}{r_a}\right) - 1]$	32.91	Ohm
4 Calculating Output current (eq. 4)		
$I_a = \frac{V_p - V_a}{R_a}$	0.03	Ampere
5 Total net Anode mass (eq. 5)		
$M_a = \frac{I_c \times tf \times 8760}{\mu \times \varepsilon}$	41.50	kg
6 Calculating Min. Number of Anodes due to Mass (eq. 6)		
$N_{min,m} = \frac{M_a}{m_a}$	3	pcs
7 Calculating Min. Number of Anodes due to Current (eq. 7)		
$N_{min,i} = \frac{I_c}{I_a}$	8	pcs
8 Selected Number of Anodes (eq. 8)		
$N = \text{Max}\ [N_{min,m}, N_{min,i}]$	8	pcs
9 Individual Anode Current Capacity (eq. 9)		
$C_a = m_a \times \varepsilon \times \mu$	13,558	Ah
10 Crosscheck to DNV RP B401 (eq. 10)		
$N \times C_a$	109,794	
$I_c \times tf \times 8760$	38,798	
$N \times C_a \geq I_c \times tf \times 8760$	**OK**	

Note : This calculation shall be confirmed by VENDOR during detail engineering